

;PAPER TITLE_i: ξ-Coupling Closure between PAPER__ia_i and PAPER__ib_i, v5.78

Star-Magic UQFF
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Abstract

This paper derives the ξ-coupling between the previously closed sectors PAPER__ia_i (*short title a_i*) and PAPER__ib_i (*short title b_i*). The coupling is required to be stationary ($\partial\xi/\partial p = 0$) at the canonical operating point in `dpm_vacuum_manifold.py` v3.0, and we verify the resulting joint prediction against ;observational anchor_i.

1 Parent Closures

PAPER__ia_i ;short title_i, closure $C_a = \langle value \rangle$, $z_a = \langle z \rangle$.

PAPER__ib_i ;short title_i, closure $C_b = \langle value \rangle$, $z_b = \langle z \rangle$.

2 Coupling Definition

$$\xi_{ab} = w_a C_a + w_b C_b + \kappa_{ab} C_a C_b \quad (1)$$

with weights w_a, w_b derived from the canonical $|SO(5)| = 10$ ratio and κ_{ab} proportional to F_{TRZ}^n where $n = \langle n \rangle$ is fixed by dimensional analysis.

3 Stationarity Constraint

At the canonical operating point,

$$\left. \frac{\partial \xi_{ab}}{\partial p} \right|_{p=p_\star} = 0 \quad (2)$$

which fixes ;which canonical parameter or relationship_i.

4 Joint Prediction

Observable	Predicted (joint)	Observed (cite)
jobs _i	;value _i	;value ± error _i

$\varepsilon = \langle Z \rangle \%$, $z = \langle z \rangle$.

5 Cross-references

- Parent `PAPER_iai`, parent `PAPER_ibi`.
- Ledger row: `xi_<a>_<b>` in `master_closures.csv`.

References

- [1] `iDOI`/arXiv for the joint observational anchor_i.
- [2] `PAPER_iai` (Star-Magic).
- [3] `PAPER_ibi` (Star-Magic).